

Functional Annotation Report: cbcX (PP_0296, UniProt Q88R38)

Pseudomonas putida KT2440 — Choline substrate-binding protein of the Cbc ABC transporter

1. Summary (Answer to the Research Question)

cbcX / PP_0296 (Q88R38) encodes the periplasmic, choline-specific substrate-binding protein (SBP) of the Cbc (choline/betaine/carnitine) ATP-binding-cassette (ABC) import system. Its primary molecular function is to capture the quaternary-ammonium compound **choline** from the periplasm with high affinity (and glycine betaine with lower affinity) and deliver it to the membrane-embedded **CbcWV** permease/ATPase core for ATP-driven uptake into the cytoplasm. It is **not itself a catalytic enzyme or a membrane transporter**; despite the EC 3.6.3.32 tag inherited from the transport reaction of the whole complex, CbcX is the recognition/receptor subunit. The imported choline is subsequently oxidized (BetAB) to the osmoprotectant **glycine betaine**, supporting osmotic-stress tolerance, and can also be catabolized as a **carbon/nitrogen source**. CbcX carries out its function in the **periplasm** (it bears a cleavable Sec signal peptide and adopts the class-II periplasmic-binding-protein / "Venus-flytrap" fold).

Identity is confirmed: the UniProt description ("Choline / betaine / carnitine ABC transporter-substrate binding protein"), gene symbol (cbcX), domains (OpuAC/PF04069; IPR017783 ABC_choline_sub-bd), and *P. putida* KT2440 genomic context all match the experimentally characterized Cbc system of *Pseudomonas* and its orthologs. There is no gene-symbol ambiguity.

2. Gene/Protein Identity Verification

Attribute	Value	Source
UniProt accession	Q88R38	UniProt
Gene symbol / locus	cbcX / PP_0296	UniProt, KEGG ppu:PP_0296
Organism	<i>Pseudomonas putida</i> KT2440	UniProt
Length	315 aa	UniProt
Signal peptide	residues 1–21 (Sec; cleaved), mature chain 22–315	UniProt
Domain	ABC-type glycine betaine transport SBP, res. 29–284	UniProt
Pfam / InterPro	PF04069 (OpuAC); IPR017783, IPR007210	UniProt/InterPro
Ortholog group	eggNOG COG2113; KEGG K02002	UniProt/KEGG
Fold	Periplasmic binding protein-like II (SSF53850; Gene3D 3.40.190.10/100)	UniProt
Genomic neighbour	PP_0295 = CbcW permease (K02001)	KEGG

Conclusion: All annotations converge on a single, unambiguous assignment. The gene is correctly identified; the symbol is not confused with any unrelated "cbcX."

3. Primary Function and Substrate Specificity

CbcX is the periplasmic **substrate-binding protein (SBP)** — the receptor subunit — of a Type I ABC importer. Its direct experimental characterization comes from the orthologous Cbc system of *Pseudomonas syringae*/*P. aeruginosa*, where the SBP encoded by the **cbcXWV** operon binds:

- **Choline** with high affinity — $K_m \approx 2.6 \mu\text{M}$ (preferred substrate)
- **Glycine betaine** with lower affinity — $K_m \approx 24.2 \mu\text{M}$

Critically, **CbcXWV-mediated betaine uptake does not occur in the presence of choline**, establishing choline as the physiologically dominant substrate of CbcX (Chen et al. 2010,

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The *P. putida* KT2440 protein (PP_0296) is the direct ortholog and carries the identical annotation and KEGG ortholog (K02002).

Within the transporter, CbcX supplies **specificity**: the shared **CbcWV** core also partners with dedicated SBPs for other quaternary amines — **BetX** (betaine-specific, $K_m \approx 0.6 \mu\text{M}$) and **CaiX** (carnitine-specific, $K_m \approx 24 \mu\text{M}$) (Chen et al. 2010). Short-chain acylcarnitines and free carnitine are imported through the **CaiX–CbcWV** assembly (Meadows et al. 2018,

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independently confirming CbcWV as a common translocation core served by interchangeable binding proteins. **CbcX is therefore the choline-recognition module of this modular importer.**

Note on EC 3.6.3.32: this number denotes the *choline-transporting ATPase* reaction of the intact ABC complex ($\text{ATP} + \text{H}_2\text{O} + \text{choline}_{\text{out}} \rightarrow \text{ADP} + \text{P}_i + \text{choline}_{\text{in}}$). CbcX contributes the substrate-recognition step but has no intrinsic ATPase/hydrolase activity — the ATPase resides in the CbcV nucleotide-binding subunit.

4. Structural Basis of Substrate Recognition

CbcX belongs to the **OpuAC / ProX family of class-II periplasmic binding proteins** (Pfam PF04069; SSF53850). Structural understanding derives from crystallized orthologs of the same fold:

- **ChoX** (*Sinorhizobium meliloti*), the direct ortholog of CbcX, was crystallized bound to **choline** and **acetylcholine**, plus closed-unliganded and semi-closed states — revealing the canonical two-lobe "**Venus-flytrap**" SBP architecture (Oswald et al. 2008,

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Oswald et al. 2009,

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Chen et al. reported that ChoX and CbcX share very similar binding properties.

- In the *Agrobacterium tumefaciens* **ChoXWV** ortholog, two pocket **tryptophan residues (W40, W87)** are essential for choline binding, and the model is built on *S. meliloti* ChoX; the system is a high-affinity transporter "with a preference for choline" (Aktas et al. 2011,

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- The paradigm compatible-solute SBP **ProX** (*E. coli*) shows that **cation- π interactions** between the ligand's positively charged quaternary amine and a **rectangular box of three tryptophans** are the key determinants of high-affinity binding (Schiefner et al. 2004,

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Mechanistic inference: CbcX recognizes the trimethyl-ammonium head-group of choline through cation- π interactions inside an aromatic (tryptophan) cage and closes around the ligand in a Venus-flytrap motion, priming delivery to CbcWV. This explains both the high affinity and the selectivity for choline over structural analogues.

5. Subcellular Localization

CbcX carries out its function in the **periplasm** (outside the cytoplasmic membrane):

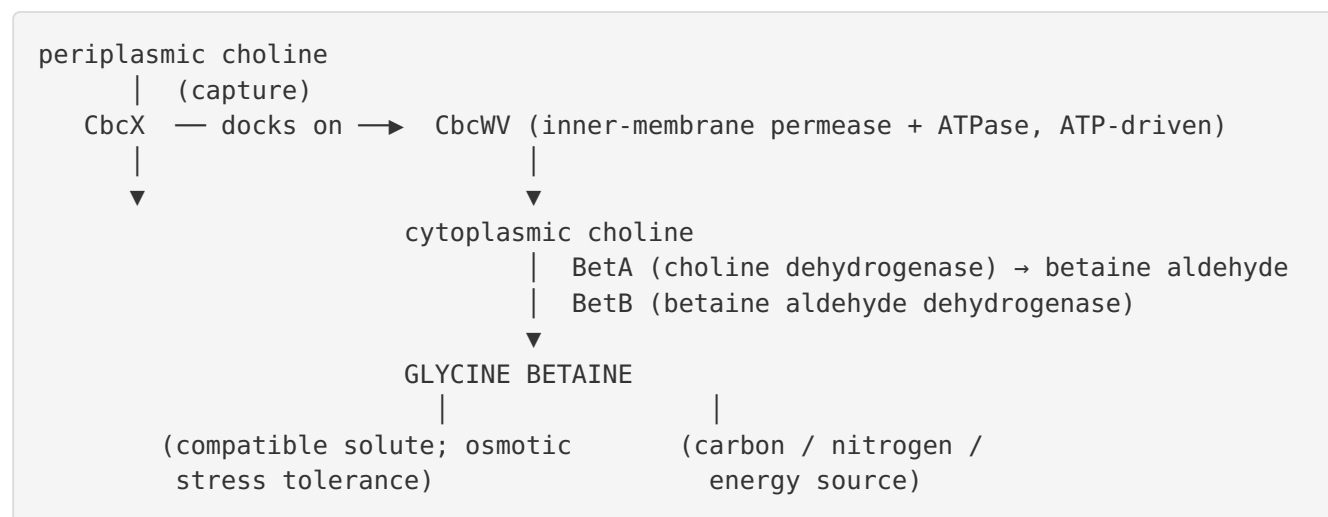
- It bears a **cleavable N-terminal Sec signal peptide (residues 1–21)** with a mature chain beginning at residue 22 — the hallmark of a **soluble periplasmic protein** (not a lipobox-anchored lipoprotein). This contrasts with the *Bacillus subtilis* ortholog OpuAC, which is a membrane-anchored lipoprotein because Gram-positive SBPs must be tethered (Kempf et al. 1997,

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- Chen et al. (2010) explicitly describe the Cbc system as using "multiple **periplasmic** substrate-binding proteins."
- The mature protein diffuses in the periplasmic space, scavenges choline that crosses the outer membrane, and docks onto the periplasmic face of the CbcWV inner-membrane complex to release ligand for translocation.

6. Biological Process / Pathway Context

CbcX is the **entry gatekeeper** of a quaternary-ammonium-compound assimilation and osmoadaptation module in *P. putida*:



- In *P. putida* KT2440, the **betBA** genes convert imported choline into the highly effective compatible solute **glycine betaine**, which is required both for endurance to high salt and for use of choline-derived compounds as a **carbon or nitrogen source** (Galvão et al. 2006,

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- The Cbc/Cho ABC system is the **high-affinity** route for choline acquisition (Aktas et al. 2011), complementing low-affinity BCCT-family secondary transporters such as BetT (Chen & Beattie 2008,

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- Because CbcX occupancy by choline blocks CbcWV-mediated uptake of other ligands (Chen et al. 2010), CbcX effectively prioritizes choline import when it is available.

This pathway is central to how *P. putida* — a soil and plant-associated bacterium living in environments rich in plant-derived choline — obtains osmoprotectants and nutrients.

7. Hypotheses: Supported and Refuted

Supported: - **H1 — CbcX is the periplasmic choline-specific SBP of an ABC importer.** Supported by direct biochemistry of the ortholog (K_m choline 2.6 μM; Chen et al. 2010), matching annotation, and genomic adjacency to the CbcW permease. - **H2 — CbcX acts in the periplasm.** Supported by the cleavable Sec signal peptide (UniProt) and the "periplasmic SBP" description (Chen et al. 2010). - **H3 — Recognition is via a Venus-flytrap aromatic (Trp) cage / cation- π mechanism.** Supported by crystal structures of ChoX and ProX and mutagenesis of essential tryptophans (PMIDs 18779321, 19642870, 21803998, 14612446). - **H4 — CbcX feeds osmoprotection and choline catabolism.** Supported by BetAB pathway data in *P. putida* (Galvão et al. 2006). - **H5 — CbcX is one interchangeable module of a shared CbcWV core.** Supported by BetX/CaiX partnership and CaiX–CbcWV acylcarnitine import (Chen et al. 2010; Meadows et al. 2018).

Refuted / corrected: - **CbcX is itself an ATPase/enzyme (as the EC 3.6.3.32 tag might suggest).** Refuted — the EC number describes the whole-complex transport reaction; CbcX is the non-catalytic receptor subunit, and the ATPase activity belongs to CbcV. - **CbcX is a membrane-integral or lipid-anchored lipoprotein (as in Gram-positive OpuAC).** Refuted for *P. putida* — the Sec signal peptide is cleaved (no lipobox), indicating a soluble periplasmic protein.

8. Evidence Quality and Limitations

- **Strongest (experimental, precise):** ligand-binding affinities and transporter reconstitution for the *Pseudomonas* Cbc system (Chen et al. 2010); crystal structures/mutagenesis of orthologs ChoX and ProX (Oswald 2008/2009; Schiefner 2004; Aktas 2011).
- **Bioinformatic/inference:** signal peptide, domain, fold, ortholog group, and gene neighborhood for the exact *P. putida* PP_0296 protein (UniProt, KEGG, InterPro).
- **Limitation:** Direct biochemical characterization of the *P. putida* KT2440 CbcX protein itself (e.g., its own K_d for choline) has not, to my knowledge, been published; the functional assignment rests on very close orthology (identical operon architecture, domain, and KEGG ortholog) to the experimentally validated *P. syringae*/*P. aeruginosa* CbcX and *S. meliloti*/*A. tumefaciens* ChoX. Given the >high sequence/architecture conservation, this inference is robust.

- **Future directions:** (i) purify *P. putida* CbcX and measure choline/betaine affinities directly; (ii) solve/predict its structure (AlphaFold model available) to confirm the tryptophan aromatic cage; (iii) test cbcX deletion phenotypes for choline uptake and osmoprotection in KT2440; (iv) clarify the exact fate/regulation (the adjacent PP_0298 AraC-family, glycine-betaine-responsive activator suggests substrate-inducible expression).

9. References (PMID)

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Database sources: UniProt (Q88R38), KEGG (ppu:PP_0296), InterPro/Pfam, eggNOG.

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